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ESP32 Audio Streaming Project Plan

Overview

ESP32-S3 device that captures mic audio and plays downlink audio with low-latency streaming to/from a host; simple on-device Wi-Fi setup, LAN discovery, and basic VAD-based wake

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Main Parts

1. Audio (Sound In & Out)

- Mic records sound > 20ms chunks
- Send sound to server
- Get sound back from server
- Speaker plays it back

2. Internet Connection

- Device connects to home Wi-Fi
- User sets up connection via phone browser (simple form)
- Device shows up on network so you can find it

3. Stay Online

- If connection drops, automatically reconnect
- Checks every 1 second that connection is working
- If device crashes, it restarts itself

4. Safety

- Encrypted connection (HTTPS-like)
- Firmware updates over internet
- Can rollback to old version if something breaks

Timeline (5-6 weeks)

Week 1: Record sound and play it back (test on device)

Week 1-2: Send sound to server, get it back, play it

Week 2: Simple setup form (phone connects to device, enters Wi-Fi password)

Week 3: Device keeps running for hours without crashing

Week 4: Encrypted connection, updates work

Week 4-5: Test everything, make sure it works 10 minutes straight

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What You Need to Check

1. **Sound quality:** Record at 16 kHz OK? (normal quality)
2. **Speed:** 250 milliseconds delay OK? (normal for internet)
3. **Speaker/Mic and other components model:** Will you give us the hardware? As you said you will provide. Please provide the preferred components so the project can start as soon.
4. **Server side:** Python or Node.js?

How We Test It Works

- Record 10 minutes straight with no problems
- Unplug Wi-Fi > device reconnects, keeps working
- Speed is under 250 milliseconds
- Can find device on network in under 2 seconds
- Firmware update works, can go back if needed

Work Hours

- Coding: ~35 hours
- Testing: ~8 hours
- Documentation: ~4 hours
- **Total: 5-6 weeks part-time (I will try to complete as soon as possible)**

Note:

The above details are only estimated so we will not depend on each and everything exactly. These are only for reference.